How to Create a New App

Checklist: from nothing to an app deployed by Jenkins at `https://magicbox.local/<app>` on every push to `main` (ADR-0016, ADR-0017). Replace `<app>` (lowercase letters, digits, hyphens; e.g. `orders-api`). `controller$` = on your workstation, in the `magicbox` repository.

0. Prerequisites (once)

- [ ] MagicBox running: `make deploy`, `make stacks`.
- [ ] Jenkins with credentials `github` and `portainer-api-token` (installation guide 4.1): `make ci-cd`.
- [ ] API gateway: `make api-gateway` (installation guide 4.2).
- [ ] `stacks_jenkins_github_user` set in `inventory/hosts.yml`.
- [ ] GitHub token of Jenkins with access to **All repositories** (Contents: Read-only, Commit statuses: Read and write), so new repositories need no token change. Otherwise add each new repository to the token.
- [ ] GitHub CLI logged in, for `--create-repo`: `gh auth status`.

1. Create the app

```
controller$ uv run tools/new_app.py <app> --create-repo
```

- [ ] Output ends with `Registered <app>: port 180xx` and the push to `https://github.com/<github-user>/<app>`.

What it did: project `../<app>` from `templates/python-fastapi-app` (renamed, `uv lock`, first commit), private GitHub repository with `main` pushed, entry in the apps registry `inventory/group_vars/magicbox/apps.yml` (first free port). Without `--create-repo`: create the private repository on GitHub, then `git -C ../<app> push -u origin main`.

Existing repository (already pushed, with the template's `Jenkinsfile` and `deploy/`): add the entry by hand instead (template: `inventory/apps.example.yml`). The gateway route `/<app>` is derived from `name` and `port`; there is no separate route setting.

```
magicbox_apps:
  - name: <app>          # job, stack, container, path /<app>
    port: 180xx          # unique, 18000-18999
    repo: https://github.com/<github-user>/<repository>
```

- [ ] Everything the pipeline needs is committed and pushed to `main`: Jenkins builds from GitHub, not from the local folder.

2. Register it on MagicBox

```
controller$ make apps
```

- [ ] Ends with `failed=0`; the report lists `<app>: registered, not deployed yet`.
- [ ] Jenkins (`https://magicbox.local/jenkins/`) shows the job `<app>`; its first scan starts the build of `main`.

3. First deployment

- [ ] Build `<app> > main > #1`: Prepare, Lint, Test, Build, Deploy, Verify green (the first build is slower: base image pull).
- [ ] Deploy log: `Created stack '<app>'`.
- [ ] `https://magicbox.local/<app>/health` answers `{"status":"ok"}`.
- [ ] Optional: `make apps` reports `<app>: deployed`.

4. Develop

```
controller$ cd ../<app>
controller$ uv sync && uv run uvicorn app.main:app --reload    # http://127.0.0.1:8000/
controller$ uv run ruff check . && uv run ruff format --check . && uv run pytest
```

- [ ] Write routes in `app/` and a unit test for each in `tests/`.
- [ ] Routes are served from `/` (Kong strips `/<app>`); use relative links.
- [ ] Push to `main`: deployed within 5 minutes. Other branches: Lint and Test only. A failing lint or test stops before the deploy.
- [ ] Do not edit `Jenkinsfile` or `deploy/`: they are generic. Container port other than 8000: change it in `Dockerfile` and `deploy/stack.yml`.
- [ ] `README.md` of `magicbox`, table "Addresses": add the app URL.

5. Remove an app

- [ ] Delete its entry from `inventory/group_vars/magicbox/apps.yml`, then `make apps` (removes the route and the port).

- [ ] Jenkins: delete the job `<app>`. Portainer: delete the stack `<app>`.

6. Test an app again from scratch

Keep the registry entry; only the job and the stack are reset.

- [ ] Portainer (`https://magicbox.local:9443/`), environment `homelab`: delete the stack `<app>` (removes the container).
- [ ] Jenkins: delete the job `<app>` (with its builds).
- [ ] `controller$ make apps`: recreates the job and starts its first scan, even when the registry is unchanged.
- [ ] Steps 2 and 3 again; Deploy log `Created stack '<app>'` (without the stack reset: `Updated stack`).

7. Troubleshooting

Symptom	Fix
new_app.py: "set stacks_jenkins_github_user"	Set it in inventory/hosts.yml or pass --repo
new_app.py: "already exists"	Pick another name or delete the folder
gh fails	gh auth login, then create the repository and push by hand (step 1)
make apps: "run make ci-cd first"	Jenkins predates the apps support: make ci-cd
make apps: assertion on <app>	Name or port duplicated or reserved, or bad repo URL
Job scan: no branches / 401	GitHub token cannot read the repository: prerequisite 0
Lint or Test fails	Run step 4 commands locally; nothing is deployed
Deploy: HTTP 401	vault_jenkins_portainer_token (installation guide 4.1), make ci-cd
Deploy: "not in the MagicBox apps registry"	Run make apps
Verify fails	HEALTHCHECK in Dockerfile; sudo docker logs <app> on MagicBox
/<app> returns 404	make apps not run since the app was added
/<app> returns 502	Not deployed yet: check the Jenkins build